

Revision Notes for Class 7 Science Chapter 7 Heat Transfer in Nature

Heat transfer is a key concept in nature and is responsible for the differences in temperature across different regions. For example, places closer to the equator are usually warmer due to the Sun's direct rays, while areas farther away or at higher altitudes remain colder. The Sun is the Earth's main source of heat and energy, influencing climate and weather patterns around the world. Understanding how heat moves helps us explain many daily observations, from why cooking pans are usually made of metal, to how houses can be kept cool or warm in different climates.

Conductors and Insulators of Heat

Some materials, like metals, are good conductors of heat, which means they allow heat to pass through easily. That's why utensils for cooking are mostly made from metals. In contrast, materials like wood, glass, clay, and porcelain do not conduct heat well. These are called poor conductors or insulators. That's why hot drinks stay warmer for longer in porcelain or clay cups. Woollen clothes help keep us warm in winter because wool traps air, which is also a poor conductor of heat, preventing our body heat from escaping easily.

- **Good conductors:** Metals like steel, aluminium, copper
- **Poor conductors:** Wood, air, glass, clay, wool, and porcelain

Air is a poor conductor of heat. That is why double-layered clothing or two thin blankets with air trapped between them keeps us warmer than a single thick blanket. Houses in cold regions often use thick wooden walls or hollow bricks filled with insulating materials to prevent heat loss, showing how traditional designs use knowledge of heat transfer.

Methods of Heat Transfer

Heat moves from one object or place to another by three main processes: conduction, convection, and radiation.

- **Conduction:** This happens mainly in solids. Heat is transferred from the hotter part to the colder part without any movement of the material itself. For example, when a metal rod is heated at one end, the other end becomes hot after some time.
- **Convection:** This takes place in liquids and gases. Heat transfer happens due to the movement of the particles themselves. For example, when water is heated in a pan, hot water rises while cooler water moves down to take its place, creating a circular motion.
- **Radiation:** This is the transfer of heat without any medium, directly from a hot object to its surroundings. The warmth you feel from the sunlight or a fire is due to radiation.

A well-known activity shows conduction using a metal strip with pins attached using wax. When one end is heated, pins fall in order starting from the end nearest the heat, showing how heat travels through the material.

Land and Sea Breeze: Example of Convection

Convection explains many natural phenomena, such as land and sea breezes along coastal areas. During the day, land heats up faster than water, causing air above the land to rise and cool air from the sea moves in to take its place—this is called sea breeze. At night, land cools down faster than water, so warm air over the sea rises and cool air from the land flows towards the sea, creating a land breeze.

Temperature Change in Soil and Water

Time (min)	Soil (°C)	Water (°C)
0	(Initial)	(Initial)

5	(↑ faster)	(↑ slower)
10		
15		
20	(Soil hottest)	(Water cooler than soil)

The table shows that soil heats and cools much faster than water, affecting wind directions along the coast.

Radiation and Daily Life Examples

Radiation allows heat from the Sun to reach the Earth. All warm objects, like hot utensils or a fireplace, also lose heat by radiation. Darker objects absorb more heat, so wearing dark clothes in winter helps you feel warmer, while light-coloured clothes reflect more heat and are comfortable in summer.

Whenever water is heated in a metal pan, all three types of heat transfer work together: conduction from the flame to the pan, convection within the water, and radiation out from the hot surfaces.

The Water Cycle: Nature's Way of Reusing Water

Heat from the Sun not only warms the Earth but also plays a key role in the water cycle. Water evaporates from oceans, lakes, rivers, and even from plants and soil. This water vapour rises, cools to form clouds, and then falls as rain, snow, or hail—a process called precipitation. The entire journey, from evaporation to precipitation and return to water bodies, is known as the water cycle.

Some rainwater seeps into the ground through a process called infiltration. It fills spaces between rocks and soil below the Earth's surface, becoming groundwater. The underground layers that store this water are called aquifers. The ease with which water

moves through soil depends on particle size; water passes quickly through gravel, more slowly through sand, and very slowly through clay.

- **Infiltration:** water moves down through soil and rocks into the ground
- **Aquifers:** underground layers storing groundwater

Due to overuse of groundwater and less infiltration in cities, water tables are falling. Solutions include rainwater harvesting and recharge pits to help restore groundwater.

Traditional Wisdom and Innovative Solutions

People adapt their homes and lifestyles according to knowledge of heat transfer. Himalayan houses may have thick wooden walls filled with insulating materials like mud and cow dung, while homes in hot regions use hollow bricks to keep interiors cool. In Ladakh, people create “ice stupas” during winter: artificial ice structures that slowly melt to provide water in summer.

Even simple devices like the “bukhari”—a traditional room heater in the Himalayas—show all three heat transfer modes in action and can be used both for heating and cooking.

Key Takeaways

- Heat moves by conduction in solids, convection in liquids and gases, and radiation without medium.
- Good conductors include metals; poor conductors (insulators) include air, wool, clay, and wood.
- Natural phenomena like land and sea breezes are due to convection.
- Our clothing and house designs utilize knowledge of heat transfer to provide better comfort.
- The water cycle keeps water circulating on Earth, and groundwater is recharged by infiltration through soil and rocks.